

F2RPC: Fake to Real Portrait Control from a Virtual Character

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ABSTRACT

This study explores a new method for generating virtual characters in video for use in virtual and augmented reality services. We propose F2RPC as a unified framework for improving both the appearance and motion of the characters, unlike previous method focusing only on one or the other. Specifically, our F2RPC consists of two modules for image destylization (AdaGPEN) and face reenactment (PCGPEN), respectively. Experimental results show that ours successfully solves the task compared to the combination of state-of-the-art destylization and reenactment method.

Index Terms: Computing methodologies—Artificial intelligence—Computer vision—Computer vision problems; Computing methodologies—Artificial intelligence—Computer vision—Computer vision representations

1 INTRODUCTION

Creating realistic virtual human characters has been actively studied in both academia and industry due to the success of virtual and augmented reality in business sectors. To achieve realism, virtual characters must have both a photo-realistic appearance (e.g., shape and texture) and natural motion. However, previous studies have focused on only one of these aspects, rather than addressing both.

Recent efforts to create more realistic virtual characters have focused on improving their appearance via destylization (realization). However, existing methods still exhibits two major limitations; (1) structure unreality and (2) texture unreality. Consequently, they require a significant amount of manual labor to reduce the gaps between virtual and real characters, such as in the rigging and texture synthesis processes. To reduce the manual labor, some studies have used generative adversarial networks (e.g., StyleGAN2) to transfer virtual human faces to more photo-realistic versions. However, these methods can lose the unique identity of each virtual character in the process. It is not desirable in certain applications, such as creating digital clones, where retaining the unique identity is important.

Another important aspect of creating realistic virtual characters is motion. Face reenactment is to create realistic facial motions from source images. Recent methods (e.g., [5]) use StyleGAN2 generator to achieve high-resolution results. However, they are based on GAN inversion thus suffer from sensitivity to misalignment and loss of identity, well-known limitations of GAN inversion.

This paper aims at (1) developing a unified framework for both appearance destylization and face reenactment, and (2) devising new appearance destylization and face reenactment modules that overcome limitations caused by GAN inversion. We use the GPEN [4] image restoration framework to address both destylization and reenactment problems, which can preserve the identity of the virtual character better than GAN inversion-based methods.

In addition, we introduce AdaGPEN and PCGPEN for improving the appearance and motion of virtual characters. AdaGPEN addresses both structure and texture unreality by using the AdaMix

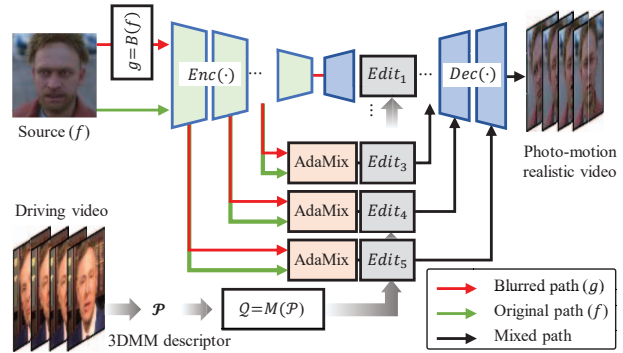


Figure 1: Architecture of F2RPC. Given a source image and 3DMM descriptors, our model generates a photo- and motion-realistic video, where it preserves the identity of the input character and faithfully follows 3D motions defined by 3D descriptors. Both AdaGPEN and PCGPEN are integrated into a single framework by sharing the encoder under the same GPEN baseline.

block. PCGPEN uses a flow-based face reenactment module that is conditioned on a 3DMM motion descriptor. It preserves the input identity and is more robust against misalignment than other methods.

Finally, we propose F2RPC, a unified framework that uses AdaGPEN and PCGPEN to create a photo-realistic virtual character video from a single character (fake) image and a motion descriptor of a source video. F2RPC combines destylization (AdaGPEN) and reenactment (PCGPEN) modules and reduces redundant operations by sharing the same encoder-decoder backbone. This approach produces high-resolution, photo-realistic and natural-looking video. Besides it avoids quality degradation due to image misalignment or loss of identity caused by GAN inversion.

2 F2RPC: FAKE TO REAL PORTRAIT CONTROL

2.1 AdaGPEN

We develop AdaGPEN to address the limitations of existing destylization methods and GPEN. Our method can (1) maintain the identity after destylization and (2) improve both structure and texture realism at the same time. GPEN baseline is used to keep the identity of the original input. AdaGPEN exploits the low-resolution features of the degraded image to improve the structure reality and the high-resolution features of the original image to enhance the texture reality. To enforce seamless styles between low- and high-resolution features, we use AdaMix block to match the style of the high-resolution features to that of low-resolution features.

2.2 PCGPEN

We propose a portrait controllable GPEN (PCGPEN), a flow field-based face reenactment module with the GPEN baseline. We employ feature-level warping from StyleHEAT to GPEN to prevent identity loss. The warping is applied at d_{64} which is an intermediate feature with a resolution 64 of GPEN decoder $Dec(\cdot)$, and branch features $\{b_{res} | res = 64, 128, 256, 512\}$. The warped result successfully reflects the intended motion without any quality degradation. PCGPEN is capable of generating an animated character video with higher reconstruction quality than StyleHEAT.

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Table 1: **A:** Destylization results on MSSF dataset. **B:** Face reenactment performance on HDTF dataset. **C:** Virtual character destylized video generation performance ('+' indicates consecutive operation).

A	DualStyleGAN	GPEN(<i>f</i>)	GPEN(<i>g</i>)	AdaGPEN (Proposed)		
FID↓	105.5	57.2	66.3	60.1		
2AFC↑ (%)	-	18.2	-	81.8		
B	FID↓	Self-FID↓	LPIPS↓	AED↓	APD↓	CSIM↑
PIRender [1]	169.1	<u>18.72</u>	0.1779	<u>0.0866</u>	0.0298	0.8498
StyleHEAT [5]	<u>156.7</u>	46.81	0.2988	0.1008	<u>0.0280</u>	0.7587
PCGPEN (Proposed)	153.8	18.37	<u>0.1855</u>	0.0749	0.0226	0.8430
C	FID↓	AED↓	APD↓	CSIM↑		
(a) DualStyleGAN [3] + StyleHEAT [5]	114.8	0.1796	0.0418	0.2210		
(b) AdaGPEN + StyleHEAT	100.86	0.1928	<u>0.0456</u>	0.3911		
(c) AdaGPEN + PCGPEN (ours, 2-stage)	91.3	<u>0.1813</u>	0.0561	<u>0.4167</u>		
(d) F2RPC (ours, 1-stage)	84.5	<u>0.1813</u>	0.0500	0.4322		

2.3 A Unified framework: F2RPC

Given AdaGPEN and PCGPEN, we can successfully integrate them into a unified framework. F2RPC creates a photo-realistic virtual character video from two inputs, a single character image f^s and a motion descriptor $\mathcal{P}$. As depicted in Fig. 1, F2RPC first destylizes the source image f^s at the feature-level for improving both structure and texture reality using AdaGPEN. Afterwards, PCGPEN is used to apply motions by feature warping and calibration at the destylized intermediate features d_{64}^{des} and $b_{res=64,\dots,512}^{des}$ using $\mathcal{P}$. As a result, F2RPC produces a high-resolution photo- and motion-realistic video without any redundant operations compared to the cascade approach. Furthermore, our framework is free from the quality degradation caused by image misalignment or poor reconstruction by GAN inversion due to BFR task-based backbone.

While none of the existing studies explicitly solve two problems simultaneously, naïve combination of two different methods (i.e., destylizing appearance and then applying face reenactment) may be a possible solution. However, this solution has limitations in both performance and computational efficiency.

3 RESULTS

Metric and dataset We use FID and LPIPS for image quality, and average expression/pose distance (AED/APD) of generated video 3DMM descriptors and that of the target video for motion, and facial identity embedding cosine similarity (CSIM) for identity. We use HD-talking-face and VoxCeleb dataset to train the F2RPC. For the source virtual character, we employ Face Synthetics dataset.

Table 1.A depicts destylization performance where f, g are original/blurred image, respectively. GPEN-based methods outperform the StyleGAN-based method. Among them, our AdaGPEN and GPEN(f) show comparable FID. However, we find ours exhibits higher two alternative forced choice (2AFC) scores in user study. This is because that GPEN(f) only improves the texture realism while the artificial structure is unchanged. However, AdaGPEN enhances both structure and texture realism. In Table 1.B, we observe that PCGPEN achieves the best image quality and motion accuracy. Besides, it can produce high-resolution results unlike PIRender (See Sec. 4 for the resolution issue in PIRender).

Table 1.C reports quantitative results among three methods. Our method exhibits the best image quality and identity accuracy. Table 1.C (a) shows the best motion accuracy but each frame has a mostly average-face-like appearance with monotone background. More seriously, it rarely preserves the character’s identity because the average-like character is favorable to control its motion via StyleHEAT. Therefore, DualStyleGAN shows high accuracy in rendering motions by generating average-like characters, which is not useful in our target scenario. We observe consistent results in (b); the image quality improves by adopting AdaGPEN over (a) while its motion accuracy is worse than (a). AdaGPEN maintains the major identity of the image during the destylization. Then, the reenactment performance deteriorates because StyleHEAT should handle a difficult source image.

For the qualitative results of AdaGPEN, PCGPEN and F2RPC, please refer to the video. In the video, it is observed that AdaGPEN successfully enhances both texture and structure reality, and PCGPEN produces clear and accurate reenacted videos. Finally, F2RPC produces realistic and high-quality images with accurate motions while preserving the identity of virtual character.

4 DISCUSSION

We find that editing features at higher resolution than 64 leads to unstable training of warping network in PCGPEN, which hampers motion quality. In the qualitative results, we observe the unrealistic texture pattern mismatch to the original (e.g. unaligned hair strands and teeth). These artifacts are due to the texture sticking of StyleGAN2. We will improve the F2RPC using StyleGAN3 architecture which reduces texture sticking.

We compared PCGPEN to PIRender and StyleHEAT. We trained PIRender to reenact $\mathbb{R}^{512 \times 512}$ image, although it is designed to reenact $\mathbb{R}^{256 \times 256}$ images. However, we found that it shows more blurry image due to unstable training, so that the quantitative results are worse than that of $\mathbb{R}^{256 \times 256}$. Thus, we compared the $\mathbb{R}^{256 \times 256}$ resolution version.

Some recent methods aim to reenact $\mathbb{R}^{256 \times 256}$ image while our target is $\mathbb{R}^{512 \times 512}$. Wang et al. [2] used same identity in both the source image and target video, while ours uses source images of different identity and domain, which is a more challenging problem.

5 CONCLUSION

We ineptively address the problem of generating realistic virtual character videos from a single character image and a driving video. We proposed F2RPC, a unified framework that addresses three technical challenges: (1) preserving the character’s identity, (2) faithfully following the target motions, and (3) creating realistic faces.

To tackle three issues, F2RPC consisted of AdaGPEN for destylization and PCGPEN for reenactment. To retain the character’s identity in the final output, we employ GPEN to tackle both destylization and reenactment problems because of its high reconstruction quality. AdaGPEN exploits GPEN baseline with AdaMix blocks to improve both structure and texture realism. PCGPEN builds a flow field-based face reenactment framework with GPEN. F2RPC unifies both modules into a single framework. F2RPC is computationally more efficient than our 2-stage model, and also exhibits higher image quality than cascading existing solutions for destylization and reenactment (DualStyleGAN+StyleHEAT). Through intensive evaluations, we show the capability of the F2RPC to solve photo- and motion-realistic virtual character video generation task.

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